

RESEARCH INTERESTS	With 4+ years of experience in embedded systems, hardware security, and intermittent computation, my research focuses on designing, evaluating, and securing resource-constrained embedded systems. My work spans hardware/software co-design, bare-metal firmware, high-throughput embedded fuzzing, trusted execution, secure checkpointing, system-level benchmarking, and randomness under intermittent operation.		
EDUCATION	PhD, Computer Science	2023 - Present	
	<i>Virginia Tech</i>	Advisor: Dr. Matthew Hicks	
	MS, Electrical and Computer Engineering	2021 - 2023	
	<i>Virginia Tech</i>	GPA: 3.8/4.0	
	BS, Electronics and Telecommunication Engineering	2017 - 2021	
	<i>Dwarkadas J. Sangvi College of Engineering</i>	GPA: 7.8/10.0	
PROFESSIONAL EXPERIENCE	Virginia Tech Graduate Research Assistant	2022 - Present	
	• Explore the vulnerability space of ultra-low size, weight and power microcontroller trusted execution environments. • Develop high-throughput fuzzing techniques for resource-constrained bare-metal devices while preserving native hardware execution fidelity. • Design and evaluate trusted execution and context management techniques in resource-constrained environments. • Evaluate the robustness of random number generators present in intermittently powered batteryless devices. • Investigate trade offs between security and performance of checkpointing techniques in intermittently powered batteryless devices.		
	Efficient Computer Embedded Engineering Intern	Summer 2026	
	• Built and evaluated MLPerf Tiny benchmarks on E1x for official submission, porting workloads to the closed-division runner and optimizing Streaming Wake Words, achieving 74% reduction in latency. • Developed drivers for the MSD-3-A microSD reader, BD79104FV-LA ADC, and OV2640 camera. Supported EVK driver bring-up and customer camera integration. • Supported initial E1 silicon bring-up through board rework, hardware setup, board configuration, and liveness testing.		
	Arm Productivity Engineering Intern	Summer 2024	
	• Improved the Medini threat-modeling workflow for Arm's Hardware Secure Development Lifecycle and contributed to microarchitecture security assessment tooling.		
	NTPC Software Engineering Intern	Winter 2020	
	• Built a web-based file management system using JavaScript, ASP.NET MVC, and SQL Server, including database and front-end design.		
TECHNICAL SKILLS	• Embedded Systems • Intermittent Computation • Hardware Security • Hardware/software co-design • Threat Modeling & Risk Assessment • TEE: Texas Instruments (TI) MSP IP Encapsulation, Arm TrustZone • TI CCS, TI DSS, MPLAB X, Atmel Studio, Keil uVision, mspdebug, GDB • AMD Vivado Design Suite, Cadence Genus, GCC, ANSYS Medini • C, Assembly (ARM, MSP430), Verilog, Python, Javascript		

SERVICE	Graduate Teaching Assistant, CS 4264: Principles of Computer Security	Fall 2026
	External Reviewer, TDSC	Spring 2026
	Guest Lecturer, CS 4264: Principles of Computer Security	Fall 2025
PUBLICATIONS	μRNG: A Framework for Assessing Randomness in Intermittent Computing Devices. <u>Prakhar Sah</u> and Matthew Hicks. ACM/IEEE International Symposium on Computer Architecture (ISCA). June 2026. Sheriff: Arresting the Confused Deputy Undermining Ultra-low SWaP Device Trusted Execution Environments. <u>Prakhar Sah</u> and Matthew Hicks. ACM Asia Conference on Computer and Communications Security (AsiaCCS). June 2026. RIPencapsulation: Defeating IP Encapsulation on TI MSP Devices. <u>Prakhar Sah</u> and Matthew Hicks. USENIX WOOT Conference on Offensive Technologies (WOOT). August 2024. Hitchhiker's Guide to Secure Checkpointing on Energy-Harvesting Systems. <u>Prakhar Sah</u> and Matthew Hicks. International Workshop on Energy Harvesting & Energy-Neutral Sensing Systems (ENSsys). November 2023.	
RESEARCH PROJECTS	Evaluating Random Number Generators in Intermittent Computing Devices: Developed a framework for evaluating RNG security under intermittent operation and applied it to 189 commercial and academic devices. Found that 132 lack on-chip RNGs, 27 use insecure RNGs, and only 23 provide trustworthy entropy sources. Outcome: ISCA '26. Secure Enclave for Confused Deputy Threats on ULSWaP Devices: Designed a hardware/software architecture for secure storage, context management, and introspection that prevents confused-deputy attacks with <1% hardware and <3% runtime overhead. Outcome: AsiaCCS '26. Interrupt-based Side-Channel Attacks against Commercial TEEs: Developed attacks that break TI MSP IP Encapsulation by exploiting residual state across context switches and lack of call-site verification. Outcome: USENIX WOOT '24. Secure Checkpointing in Energy-Harvesting Devices: Analyzed existing secure checkpointing techniques and found that most overlook adversaries with physical access. Proposed a more practical and energy-aware adversarial model. Outcome: ENSsys '23.	
OTHER PROJECTS	Porting Cryptographic Benchmarks for Intermittent Computing Implementation and Evaluation of a Polling Server on ATmega2560 Implementation and Mitigation of Spectre Attacks on RISC-V Microarchitecture Object Detecting Autonomous Water Vehicle	
PRESENTATION & TALKS	μRNG: A Framework for Assessing Randomness in Intermittent Computing Devices. (ISCA) 2026 Sheriff: Arresting the Confused Deputy Undermining Ultra-low SWaP Device Trusted Execution Environments. (AsiaCCS) 2026 RIPencapsulation: Defeating IP Encapsulation on TI MSP Devices. (WOOT) 2024 Hitchhiker's Guide to Secure Checkpointing on Energy-Harvesting Systems. (ENSsys) 2023	